Photon Correlation Tomography

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Abstract

Derive 3D models of particle form factors from angle-resolved static and dynamic light scattering

1 Introduction

1.1 Electric Field and Intensity Correlations

Correlations among fluctuations of the (complex) scattered electric field, E, between time t and lag time τ are given by

$$\begin{aligned}\langle E(t)\,E^*(t)\,E(t+\tau)\,E^*(t+\tau)\rangle_t = \\ \langle|E(t)|^2\rangle_t\langle|E(t+\tau)|^2\rangle_t + |\langle E(t)\,E^*(t+\tau)\rangle_t|^2 + \langle\delta\,|E(t)|^2\,\delta\,|E(t+\tau)|^2\rangle_t\end{aligned} \tag{1}$$

where E^* is the conjugate of the complex electric field and the average $\langle\cdot\rangle$ is taken over all times t. Field fluctuations can be quantified with coherent optical scattering via the measured intensity, $I = |E|^2$. Thus, the intensity correlation is

$$\langle I(t)I(t+\tau)\rangle_t = \langle|E(t)|^2\rangle_t\langle|E(t+\tau)|^2\rangle_t + |\langle E(t)\,E^*(t+\tau)\rangle_t|^2 + \langle\delta\,|E(t)|^2\,\delta\,|E(t+\tau)|^2\rangle_t \tag{2}$$

The first term on the right of Eq. 2 simplifies to

$$\langle|E(t)|^2\rangle_t\langle|E(t+\tau)|^2\rangle_t = \langle I\rangle^2 \tag{3}$$

Dividing Eq. 2 by $\langle I\rangle^2$ gives the intensity correlation function $g^{(2)}(\tau)$

$$g^{(2)}(\tau) = \frac{\langle I(t)I(t+\tau)\rangle_t}{\langle I\rangle^2} \tag{4}$$

With the field correlation function $g^{(1)}(\tau)$ defined as

$$g^{(1)}(\tau) = \frac{\langle E(t)E^*(t+\tau)\rangle_t}{\langle|E(t)|^2\rangle_t} \tag{5}$$

Eq. 4 can be written as

$$g^{(2)}(\tau) - 1 = \left|g^{(1)}(\tau)\right|^2 + \frac{\langle\delta\,|E(t)|^2\,\delta\,|E(t+\tau)|^2\rangle_t}{\langle I\rangle_t^2} \tag{6}$$

We measure $g^{(2)}(\tau)$ and model $g^{(1)}(\tau)$. Residual intensity is attributed to higher-order correlations from the last term in Eq. 6.

1.2 Intensity Correlation Measurement

A decoherence time Γ is given by

$$\Gamma = q^2 D \tag{7}$$

where D is the diffusivity and q is the scattering vector

$$q = \frac{4\pi n \sin(\theta/2)}{\lambda} \tag{8}$$

where n is the refractive index, θ is the scattering angle relative to the incident direction, and λ is the wavelength. Diffusivity is given by the Einstein relation

$$D = \mu k_B T \tag{9}$$

where k_B is the Boltzmann constant, T is the absolute temperature, and μ is the mobility. Coherence decays exponentially as

$$g^{(1)}(q, \tau) = e^{-k_b T q^2 \mu \tau} \tag{10}$$

$\beta \in [0, 1]$ is the spatial coherence of the optical system.

Scatterers that produce a distribution of intensities, $I(q, R)$, which depend on their shape and size, decohere through a mixture of independent and correlated motion. The total intensity correlation function is the sum

$$g^{(2)}(\tau) - 1 = \beta \Big| \int I(q, R_i) e^{-k_b T q^2 \mu \tau} \Big|^2 \tag{11}$$

$$\frac{1}{2} \ln \left(\frac{g^{(2)}(\tau) - 1}{\beta} \right) = \ln \left(\Big| \sum_i I(q, R_i) e^{-k_b T q^2 \mu \tau} \Big| \right) \tag{12}$$